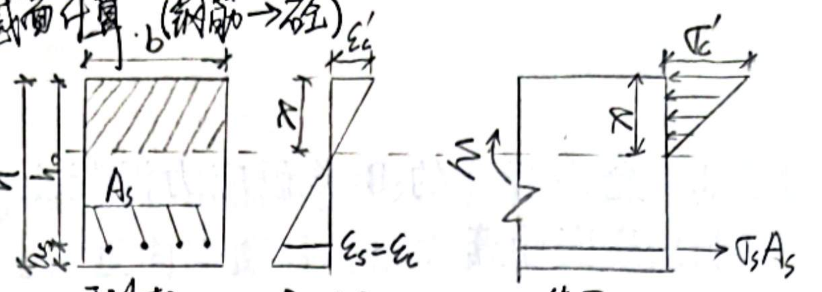
换算截面:

目的:将钢筋和砼两种材料组成的实际截面换算成一种假想材料组成的匀质截面,从而能采用材料力学公式进行截面计算.(钢筋→砼)

1.开裂的矩形截面



开裂截面 应变分布 计算图式

(1)平截面假定

$\varepsilon_c=\varepsilon_s$

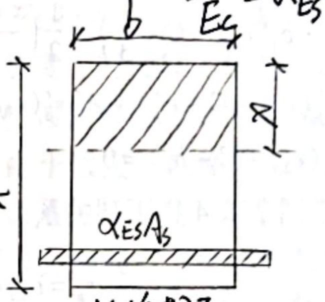
(2)受压区,压力由砼承担

$\sigma_c'=\varepsilon_c' E_c$

(3)受拉区,拉力完全由钢筋承担 $\sigma_s=\varepsilon_s E_s \Rightarrow \varepsilon_s=\frac{\sigma_s}{E_s}=\frac{\sigma_c}{E_c}\Rightarrow\sigma_c=\sigma_s\frac{E_c}{E_s}=\frac{\sigma_s}{\alpha_{Es}}$

(4)砼承担的总拉力与钢筋承担的总拉力相等. $\frac{E_s}{E_c}=\alpha_{Es}$

$\sigma_c A_c=\sigma_s A_s\Rightarrow\frac{\sigma_s}{\alpha_{Es}}A_c=\sigma_s A_s\Rightarrow A_c=\alpha_{Es}A_s$



换算截面

a.换算截面的面积 $A_{cr}=bx+\alpha_{Es}A_s$

b.换算截面的面积矩(静矩)

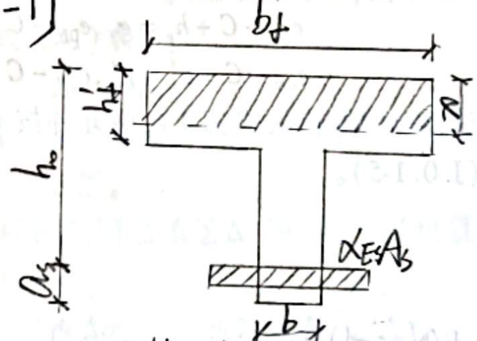
受压区 $S_{crc}=\frac{1}{2}bx^2$ 受拉区 $S_{crt}=\alpha_{Es}A_s(h_0-x)$

c.换算截面的惯性矩

$I_{cr}=\frac{1}{3}bx^3+\alpha_{Es}A_s(h_0-x)^2$

d.受压区高度x $x=\frac{\alpha_{Es}A_s}{b}\left[\sqrt{1+\frac{2bh_0}{\alpha_{Es}A_s}}-1\right]$

2.开裂的T形截面



第一类T形截面

$x=\frac{\alpha_{Es}A_s}{b_f'}\left[\sqrt{1+\frac{2b_f'h_0}{\alpha_{Es}A_s}}-1\right]$

$x\le h_f'$(第一类T形截面)

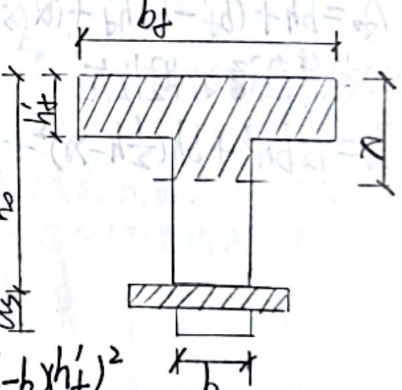
(1)换算截面的面积

$A_{cr}=b_f'x+\alpha_{Es}A_s$

(2)换算截面的面积矩(静矩)

受压区 $S_{crc}=\frac{1}{2}b_f'x^2$ 受拉区 $S_{crt}=\alpha_{Es}A_s(h_0-x)$

(3)换算截面的惯性矩 $I_{cr}=\frac{1}{3}b_f'x^3+\alpha_{Es}A_s(h_0-x)^2$



第二类T形截面

$x>h_f'$(第二类T形截面)

$x=\sqrt{A^2+B}-A$ $A=\frac{\alpha_{Es}A_s+(b_f'-b)h_f'}{b}$ $B=\frac{2\alpha_{Es}A_s h_0+(b_f'-b)(h_f')^2}{b}$

(1)换算截面的面积 $A_{cr}=bx+(b_f'-b)h_f'+\alpha_{Es}A_s$

(2)换算截面的面积矩(静矩) $S_{crc}=\frac{1}{2}bx^2+(b_f'-b)h_f'(x-\frac{h_f'}{2})$ 受压区 $S_{crt}=\alpha_{Es}A_s(h_0-x)$ 受拉区

(3)换算截面的惯性矩 $I_{cr}=\frac{b_f'x^3}{3}-\frac{(b_f'-b)(x-h_f')^3}{3}+\alpha_{Es}A_s(h_0-x)^2$

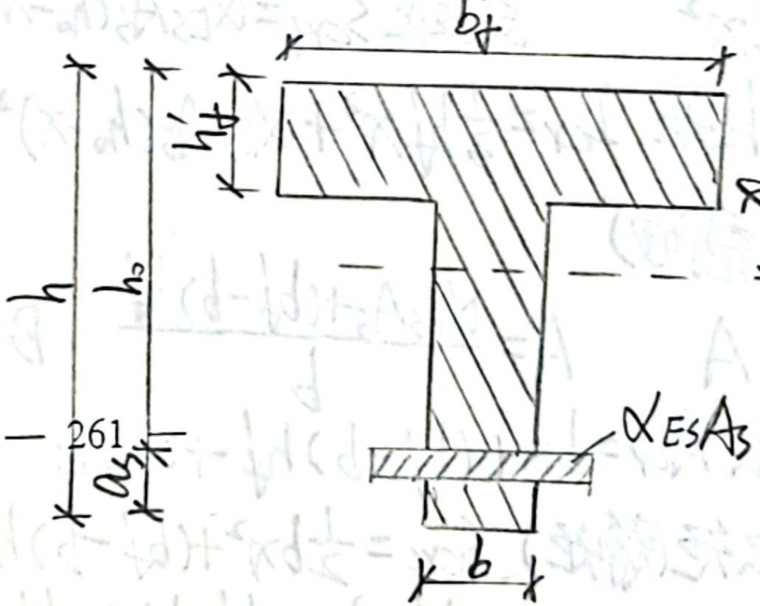
3.全截面换算截面

(1) 换算截面的面积

$$A_0 = bh + (b_f' - b)h_f' + (\alpha_{ES} - 1)A_s$$

(2) 换算截面的惯性矩

$$I_0 = \frac{1}{12}bh^3 + bh\left(\frac{1}{2}h - x\right)^2 + \frac{1}{12}(b_f' - b)(h_f')^3 + (b_f' - b)h_f'\left(\frac{h_f'}{2} - x\right)^2 + (\alpha_{ES} - 1)A_s(h_0 - x)^2$$



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